# Experimental Design: ClassSync

## Team:

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## ClassSync:

ClassSync is a browser extension that creates an integrated Outlook calendar (.ics file) for our quarters at Western. It aims to aid students in schedule management.

## Hypothesis:

1. WWU students are satisfied with the usability of ClassSync.
2. ClassSync is effective in improving WWU students' efficiency in schedule management.

**Rationale**:

- The first hypothesis tests user satisfaction with the system's design, specifically, how easy it is for users to use and interact with the system.
- The second hypothesis focuses on the system's functionality and tests if the system helps students manage their schedules quickly and correctly.

These two hypotheses allow us to evaluate the system in terms of usability and functionality. We want to prove that ClassSync is a usable system for effective schedule management.

## Subjects and sampling criteria:

**Subjects:**

1. WWU students who use ClassSync
2. WWU students who do not use ClassSync

**Sampling criteria:**

1. Must be a full-time WWU student (8~12 credits)
2. Must be enrolled at WWU for the upcoming academic year
3. Willing to fill out a follow-up survey at the end of every quarter for the academic year

**Sampling method:**

We use stratified random sampling. It divides students into groups based on class levels (freshman, sophomore, junior, senior, master's) and randomly chooses 50 students from each group who meet the sampling criteria.

**Rationale**:

- **[WWU Students]** For our subjects, we want to keep our scope to be within WWU students since this is our target audience. Our subjects are divided into those who use ClassSync and those who do not.

- **[Full-time Students]** We also want to ensure that we are recruiting full-time students who are available for the academic year. Limiting the study to full-time students allows for more reliable data. Since everyone will have a similar course load to manage, it gives us a consistent baseline for comparison. Also, since our study duration is for a year, we want to minimize the dropout rates of the study. Although there may be unexpected situations that cause students to drop out of the study, we want students to at least be expected to be available for the upcoming year before participating.
- **[Stratified Random Sampling]** Students' schedule management skills may vary with class levels, e.g., freshmen may not be as efficient in managing their times, and seniors can have a much busier schedule. This allows all WWU students in different class levels to be equally represented, and we can identify anomalies within groups.

## Variables:

**Independent:**

- Use of ClassSync

**Dependent:**

- User satisfaction on usability (SUS evaluation score of the system)
- Schedule management (self-rated survey)
    - Confidence in schedule management
    - Reliability of schedule management tool
    - Accuracy in keeping track of important dates/deadlines

**Control:**

- Student enrollment status (full-time undergraduate/graduate)
- Comfort in using online calendars

**Rationale:**

- **[Independent Variables]**The only change we make is whether a student uses ClassSync, and those who do not use ClassSync are free to use any tools. This allows a clear comparison of how schedule management skills differ between those who use the system and those who do not.
- **[Dependent Variables]**To measure one's efficiency and accuracy in schedule management, some metrics can be how they feel about schedule management, whether they have made good use of their schedule management tool, and how well they have been managing their time. These metrics give us a clear understanding of how well students are managing their schedules. We also ask users to fill out an SUS evaluation survey to understand user satisfaction and their perception of the system's usability.
- **[Control Variables]**Because ClassSync is designed specifically for WWU students, we want to only sample WWU full time students in our study. This gives a clear representation of our student population. Another control is to be comfortable in using online calendars, which avoids biases due to lack of knowledge in such technology.

## Study type:

Our Longitudinal Study will have a duration of one academic year, starting from the fall quarter and ending in the spring quarter. Half of the students use ClassSync and fill out a survey at the end of every quarter and an SUS evaluation at the end of the year. The other half continues with their current schedule management tool and fills out a survey at the end of every quarter.

**Rationale:**

- ClassSync aims to help students with schedule management throughout their academic journey at Western, so it is important to look at how it benefits students in the long run.
- Conducting a longitudinal study allows us to see if ClassSync can help promote better schedule management habits while reducing the cognitive load over time.

## Experiment Design:

We will use the between-subjects strategy, where half of the chosen students continue with their original way of schedule management and the other half uses ClassSync.

**Sampling**:

Randomly chosen 50 students from each class level (a total of 250 students). From each group, we randomly chose 25 of them to be introduced to ClassSync and the other 25 did not use it.

**Experiment**:

1. Students fill out a pre-experiment survey about schedule management.
    a. **Experimental group:**
    Students use ClassSync for the upcoming academic year
    b. **Control group:**
    Students manage their schedules freely and note down what they have used
2. At quarter-end, all students fill out the schedule management survey.
3. At academic year-end, the experimental group fills out a SUS evaluation survey for ClassSync.

**Schedule Management Survey:**

Below is a sample schedule management survey that we would ask students to fill in.

Rate each question on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree):

1. I often feel like I am disorganized with schedule management
2. I am confident with correctly keeping track of all my important dates
3. I often miss important deadlines or events
4. I think that I have good time management skills
5. I spend too much time organizing my schedule
6. My current scheduling tool is reliable and accurate
7. I often feel stressed about managing my time effectively
8. I would continue to use my current schedule management tool

**Evaluation**:

- Based on research on SUS evaluation, we will calculate the user response into a score using a SUS Calculator and assign it to grades A- F. A score above 80.3 is an A and anything below 51 is an F. Grade A indicates that the user thinks the ClassSync system is usable and learnable and F is an unusable system.

  If our first hypothesis holds, we should expect a grade of A.

- To analyze the user's schedule management behaviour, we calculate the scores for the schedule management surveys in a similar way to the SUS evaluation survey. For odd-numbered questions, we subtract user response from 5. For even-numbered questions, we subtract 1 from the user response. Adding the scores up gives us a total score out of 40. We find the average of each class-level group and compare the experiment and control group scores, and we also find a grand average of control and experimental groups as a whole.

  If our second hypothesis holds, the experimental group score should be higher than the control group score. Otherwise, we reject our hypothesis.

**Rationale**:

- **[Between-subjects Strategy]** First, it would not work if we had students participating in the study for 2 years, especially when we have seniors participating in the study. It also avoids problems like ordering effects and avoids the complexity of implementing counterbalancing methods.
- **[Schedule Management Survey]** Conducting the schedule management survey over a year is the most optimal way to quantify and measure schedule management. It allows us to get consistent feedback for identifying patterns and comparing data between experimental and control groups. It ensures that our evaluation of ClassSync's effectiveness is reliable and provides us with more room for improvement.
- **[SUS Evaluation]** SUS is an industry standard for evaluating usability, which ensures reliable and interpretable results. We also created a survey to measure functionality in terms of how well students are managing their schedules. This survey includes important metrics that represent key aspects of effective schedule management. It allows us to quantify and compare schedule management skills among the two groups. These two evaluation methods allow us to test our two hypotheses and conduct a comprehensive evaluation of ClassSync.